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DUNMAN HIGH SCHOOL Preliminary Examination Year 6

H2 BIOLOGY

9744/01

Paper 1 Multiple Choice Questions

27 September 2021

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your centre number, index number, name and class at the top of this page.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

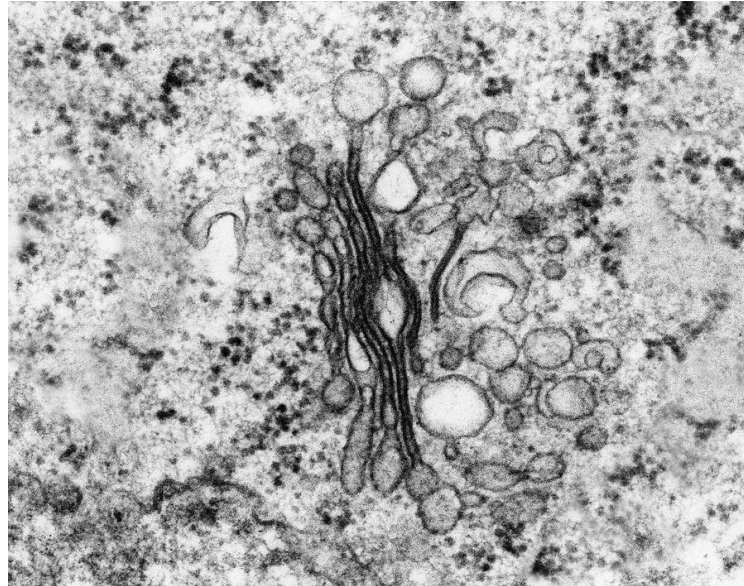
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **21** printed pages (including this cover page) and **1** blank page.

Answer **all** questions.

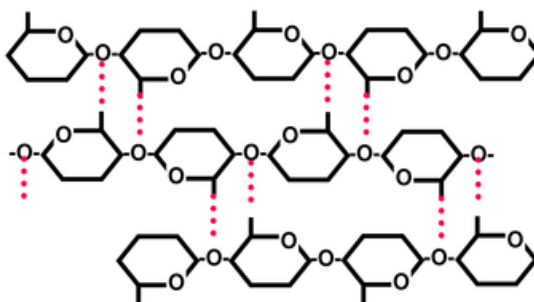
- 1 Figure below shows an electron microscopic image from part of an eukaryotic cell showing several organelles.



Which of the following is **not** a function of the organelles shown in the figure?

- A Store excess glucose to maintain cytosolic osmotic pressure
- B Chemically modify proteins with the addition of glycan chains
- C Carry transport proteins to the cell surface membrane
- D Digest ingested materials

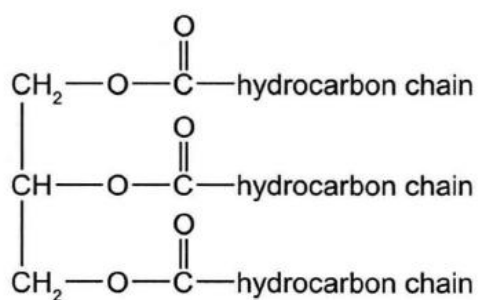
- 2 The figure below shows part of a biomolecule. The dotted lines represents hydrogen bonds.



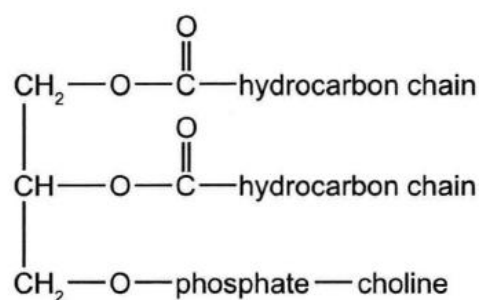
Which statement correctly describes this molecule?

- A** The three strands coils around each other to form a triple helix.
- B** This molecule is soluble in water due to limited number of groups involved in hydrogen bonding.
- C** The monomers are joined together by ester bonds.
- D** The hydrogen bonds give this molecule its high tensile strength.

3 The diagram shows the structure of two lipid molecules.



X



Y

Which row correctly matches the identification of X and Y to their description?

	Identify X	Describe X	Identify Y	Describe Y
A	triglyceride	Made by condensation reactions with the removal of 3 water molecules.	phospholipid	Amphipathic due to presence of phosphate – choline and hydrocarbon chain.
B	triglyceride	Hydrocarbon chains do not contain C=C double bonds.	phospholipid	Hydrocarbon chains contain C=C double bonds to regulate membrane fluidity.
C	phospholipid	Made by condensation reactions with the removal of 3 water molecules.	triglyceride	Hydrocarbon chains contain C=C double bonds to regulate membrane fluidity.
D	phospholipid	Hydrocarbon chains do not contain C=C double bonds.	triglyceride	Amphipathic due to presence of phosphate – choline and hydrocarbon chain.

- 4 A student was performing food tests on some solutions and he recorded his observations in the table below. However, he forgot to include headings in his table and what each row and column represented.

	Test P	Test Q	Test R
Solution A	Brick-red precipitate	Brick-red precipitate	Remained blue
Solution B	Remained blue	Brick-red precipitate	Remained blue
Solution C	Brick-red precipitate	Brick-red precipitate	Turned violet

Which statement is most likely correct?

- A All 3 solutions contain reducing sugars.
 - B Solution C contains two types of substrates.
 - C All 3 tests were performed with boiling water bath.
 - D Solution C contains starch and amylase.
- 5 During an enzyme-catalysed reaction, these events occur.
- 1 Substrate collides with enzyme
 - 2 Enzyme/substrate complex forms
 - 3 Enzyme releases product

Which row describes one possible set of processes that happen at events 1, 2 and 3?

	1	2	3
A	Conformational changes to aid temporary bonding	Conformational changes to break bonds	Bonds stressed to reduce activation energy
B	Conformational changes to aid temporary bonding	Bonds stressed to reduce activation energy	Conformational changes to break bonds
C	Conformational changes to break bonds	Conformational changes to aid temporary bonding	Bonds stressed to reduce activation energy
D	Bonds stressed to reduce activation energy	Conformational changes to break bonds	Conformational changes to aid temporary bonding

- 6 The activity of an enzyme, E, is measured under different conditions. The same amount and concentration of enzyme was used in each set up.

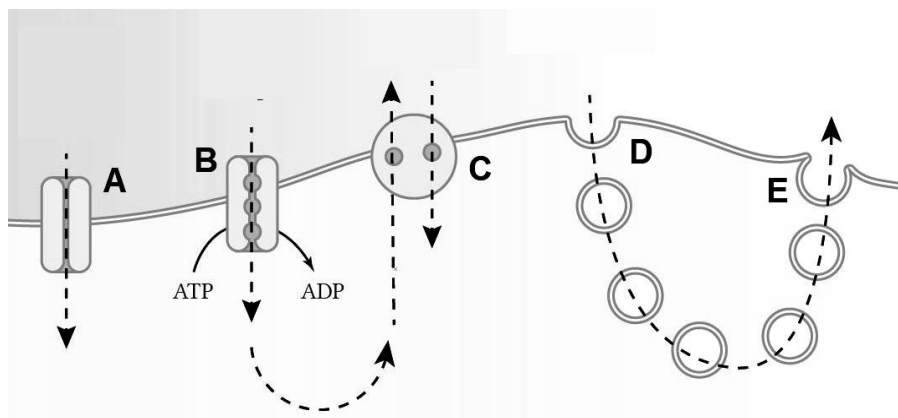
Set up	Activity of E / AU
10% Substrate, 30°C	5
10% Substrate, 40°C	10
10% Substrate, 50°C	13
10% Substrate, 1% Inhibitor, 40°C	7
20% Substrate, 1% Inhibitor, 40°C	6

Which row shows a match between a possible conclusion from the results and the supporting evidence?

	Conclusion	Supporting evidence
A	The enzyme optimum temperature is 50°C.	It is the temperature with the highest enzyme activity.
B	There is end-product inhibition.	At higher substrate concentration, the enzyme activity is lower, at the same temperature.
C	The inhibitor binds to the active site of the enzyme.	An increase in substrate concentration did not overcome the inhibition.
D	The inhibitor changes the active site of the enzyme.	Enzyme activity did not increase at higher substrate concentration.

- 7 Which statement correctly relates the structure of cholesterol to its role in the membrane?
- A** It has C=C double bonds to regulate the membrane fluidity.
 - B** It being amphipathic allows it to interact with the phospholipid molecules and serves as a molecular plug to regulate transport across the membrane.
 - C** The OH group present in its structure enables it to function as an organic solvent for transporting molecules across the membrane.
 - D** It has hydrophobic fatty acid tails that interact with the phospholipid tails, holding them close at high temperature and preventing the membrane from breaking.

- 8 The figure below shows different mechanisms of transport across membrane, labelled A to E.



Which of the following shows a correct organisation of the mechanisms according to their features?

A	Passive	Active
	A, C	B, C, D, E

B	Transport polar molecules	Transport non-polar molecules
	A, B, C	D, E

C	Transport molecules down concentration gradient	Transport molecules against concentration gradient
	A, C	B, C, D, E

D	Transport large molecules	Transport small molecules
	A, B, C	D, E

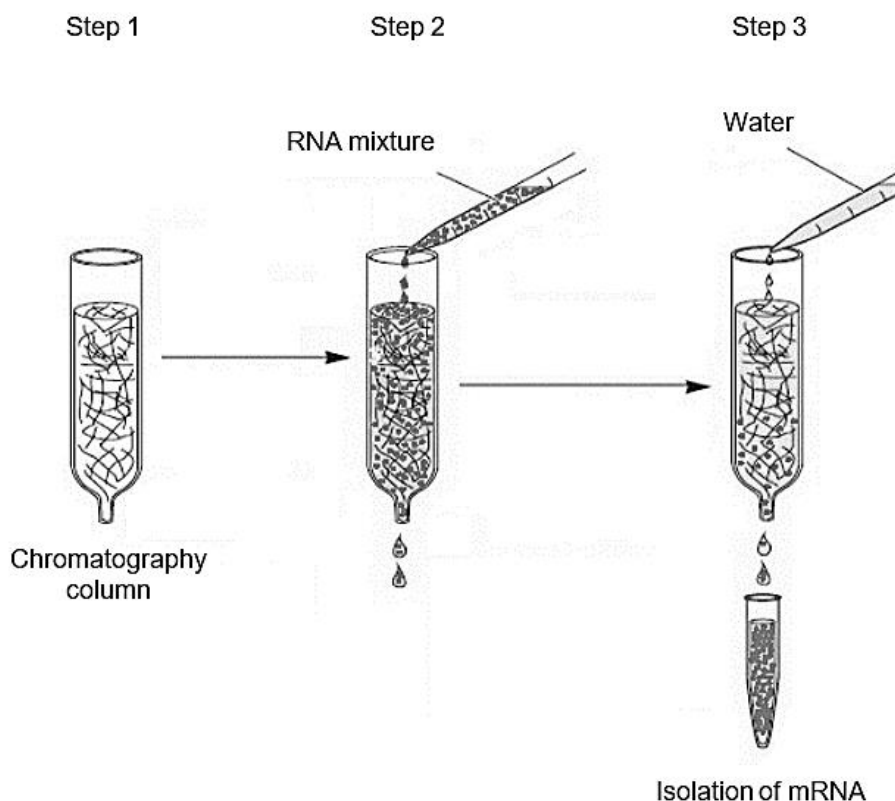
- 9** The table shows the percentages of bases in DNA from various sources.

source of DNA	adenine	guanine	thymine	cytosine
calf thymus	28.2	21.5	27.8	22.5
bull spleen	27.9	22.7	27.3	22.1
bull sperm	28.7	22.2	27.2	22.0
rat bone marrow	28.6	21.4	28.4	21.5
yeast	31.3	18.7	32.9	17.1

What is a valid deduction from the data?

- A** All cells from the same species have about the same amount of DNA.
 - B** The four bases show complementary base pairing.
 - C** The ratio of purine to pyrimidine in yeast cells is higher than animal cells.
 - D** DNA structure is linear in both yeast and animal cells.
- 10** Which statements are true for both DNA replication and PCR?
- 1 RNA primer is required
 - 2 Double stranded DNA must be separated before replication can begin
 - 3 The DNA template is read in the 5' to 3' direction
 - 4 dNTPs are required for synthesis of DNA
- A** 2 and 4
 - B** 1, 2 and 4
 - C** 2, 3 and 4
 - D** 1, 2, 3 and 4

- 11** The isolation of mature mRNA can be done by passing an RNA mixture obtained from homogenised tissue through a chromatography column. The procedure of the mRNA isolation is as follows:



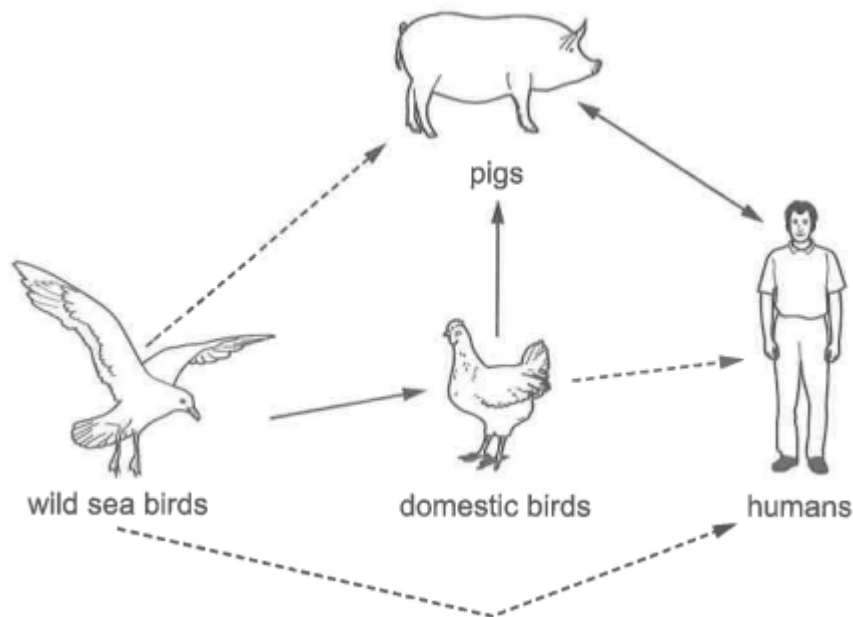
Step 1	Set up a chromatography column which contains short lengths of uracil nucleotides attached to a solid support medium.
Step 2	Add RNA mixture to the chromatography column. RNA that do not hybridise with the uracil nucleotides will pass through and leave the column.
Step 3	Add water to the chromatography column to remove and isolate the hybridised mRNA.

Which of the following statements is **not** true?

- A** rRNA and tRNA molecules could pass through and leave the column in Step 2.
- B** Complementary base pairing is responsible for the attachment of RNA to the solid support medium.
- C** The isolated mRNA in Step 3 do not contain introns.
- D** The RNA that attached to the chromatography column lack poly-(A) tails.

12 Various strains of influenza virus A can infect birds, pigs and humans.

The figure below shows the possible interspecies transmission that may take place. High risk of transmission of influenza A virus between species is shown by solid arrows. A dotted arrow shows low risk of transmission.

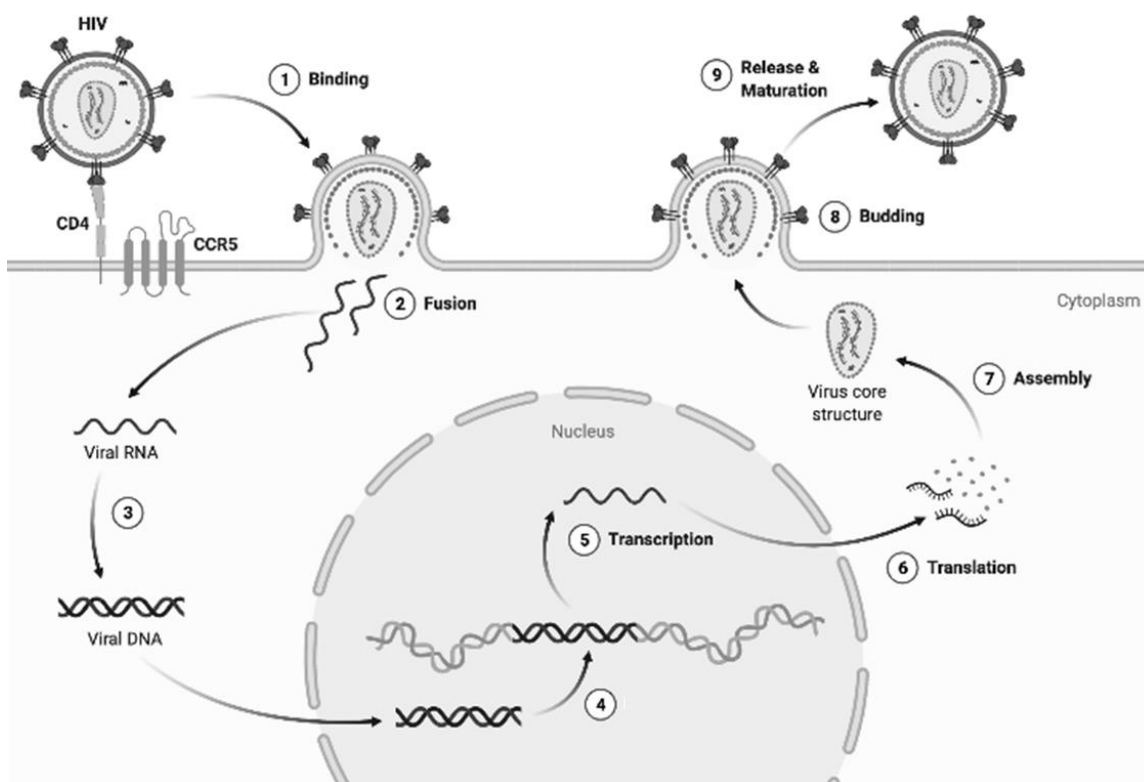


Which of the following statements correctly describes a likely situation that will negatively impact human health?

- 1 Antigenic shift due to co-infection of pig by domestic bird flu strain and human strain.
- 2 Antigenic shift due to co-infection of domestic birds with pig flu strain and human flu strain.
- 3 Antigenic drift in wild sea birds flu strain
- 4 Antigenic drift in human flu strain

- A** 1 and 2
B 1 and 4
C 1, 2 and 4
D 1, 2, 3 and 4

13 The figure below shows the HIV replication cycle in 9 stages.



Which of the following statements are true?

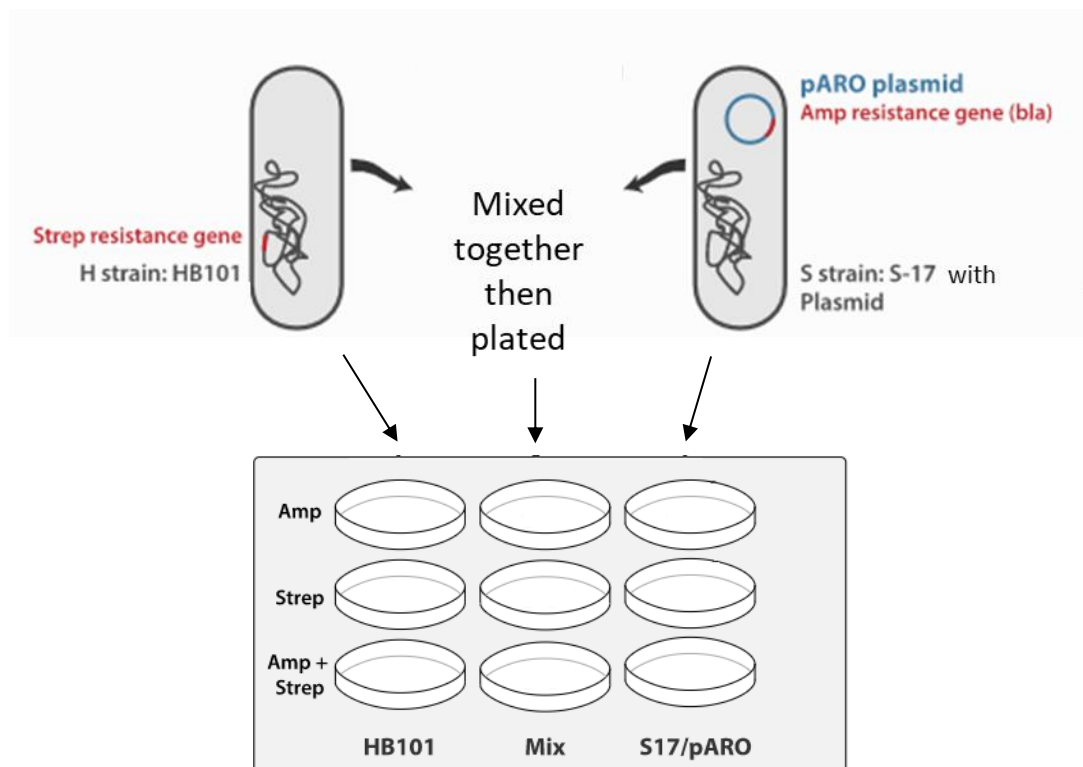
- 1 Steps 3, 4 and 5 are carried out by viral enzymes
- 2 Steps 6 uses host ribosomes
- 3 Between step 6 and step 7, a viral enzyme is required to form functional proteins
- 4 Step 1 involves viral glycoprotein hemagglutinin binding to the host cell's receptor

- A** 1 and 2
B 2 and 3
C 1, 2 and 3
D 2, 3 and 4

14 In an experiment to study conjugation between bacteria, scientists used 2 strains of *E. coli*.

- H strain – HB101 with no plasmid. The bacterial chromosome contains a gene providing resistance to streptomycin.
- S strain – S17 containing the F plasmid pARO with a resistance gene to ampicillin.

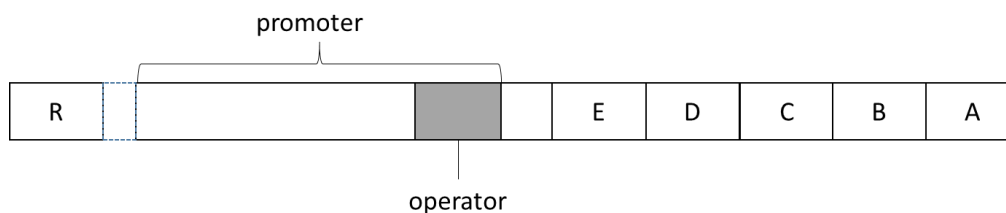
The H and S strains were mixed together. The strains H, S and the mix were then plated on various culture media as shown in the figure below.



What would the results of the experiment look like for each of these plates if conjugation was successful?

	Amp	Strep	Amp + Strep
A	Only H strain grew	Only S strain grew	H, S and mix grew
B	S strain and mix grew	H strain and mix grew	Only mix grew
C	S strain and mix grew	H strain and mix grew	H, S and mix grew
D	Only S strain grew	Only H strain grew	Only mix grew

- 15 The *trp* operon is shown in the diagram below. Upstream of the operon, *trp* R gene codes for the repressor protein that binds to the operator.



Three bacterial mutants were identified and each had a mutation to a component of the *trp* operon.

The table below shows the expression profile of the mutants.

Mutant 1:	continuous transcription of enzymes E, D, C, B and A
Mutant 2:	no transcription of enzymes E, D, C, B and A
Mutant 3:	normal regulation of transcription of enzymes, but enzyme D is non-functional.

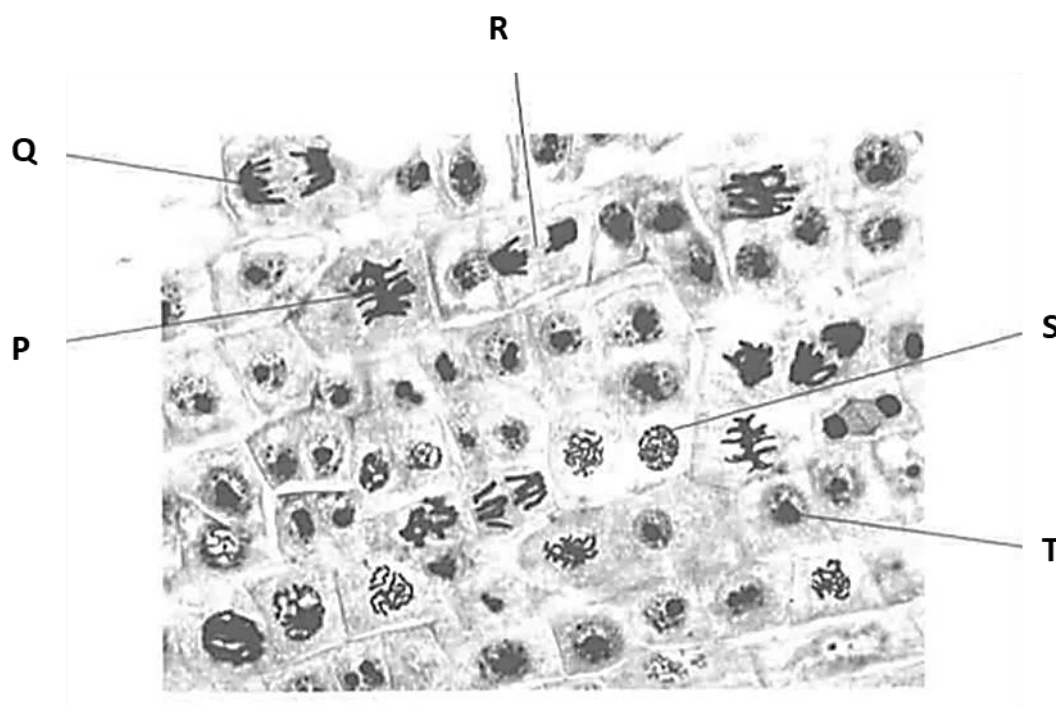
Which of the following correctly describes the possible mutations for each mutant?

	Mutant 1	Mutant 2	Mutant 3
A	Loss of function mutation to the <i>trp</i> R gene	Mutation to the operator such that repressor cannot bind	base-pair substitution in <i>trp</i> D gene
B	Mutation to the promoter such that RNA pol cannot bind	Gain in function mutation to the <i>trp</i> R gene	base addition in <i>trp</i> D gene
C	Gain in function mutation to the <i>trp</i> R gene	Loss of function mutation to the <i>trp</i> R gene	base deletion in <i>trp</i> D gene
D	Mutation to operator such that repressor cannot bind	Gain in function mutation to the <i>trp</i> R gene	base-pair substitution in <i>trp</i> D gene

- 16** In the human genome, a single gene on chromosome 11 codes for two different peptide hormones. One has 32 amino acids and plays a role in calcium metabolism. The other has 37 amino acids and stimulates dilation of arteries.

At which stage is there a difference that determines which peptide is made?

- A** Chromatin level
 - B** Between DNA and pre-mRNA
 - C** Between pre-mRNA and mRNA
 - D** Between mRNA and polypeptide
- 17** The photomicrograph shows a plant tissue sample prepared from a growing root tip.



Identify which type of division is taking place in this tissue and identify which labelled cell shows the next stage after cell S.

- A** Mitosis, cell Q
- B** Mitosis, cell P
- C** Meiosis, cell T
- D** Meiosis, cell R

18 Which statement does **not** explain how cancer arises?

- A** Mutations resulting in the loss of contact inhibition.
- B** Increased expression of genes that promotes cell division.
- C** Shortening of telomeric length with age.
- D** Error during cell division that resulted in duplication of cell cycle promoting genes.

19 A group of scientists had recently used gene editing technique to delete a defective gene that is responsible for a fatal heart condition, in human embryonic stem cells.

What could be a possible ethical concern?

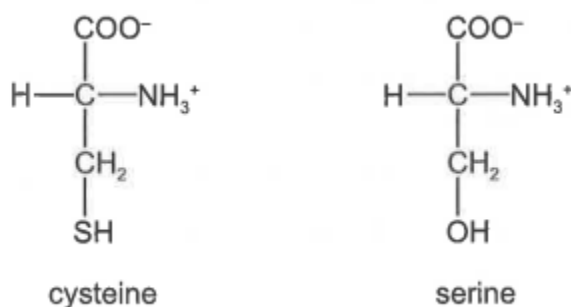
- A** The use of embryonic stem cells might result in the eventual death of an individual.
- B** The use of embryonic stem cells suggests the prevention of the embryonic stem cells from using the cells for reproduction.
- C** The successful use of gene editing techniques suggests the possibilities of creating designer babies.
- D** The gene editing technique is new and expensive, available only to affluent group of scientists.

- 20** Which options correctly show the type of mutations that can occur in each of these phases of the cell cycle and whether the mutation will be passed down in the gametes?

	Phase of cell cycle	Type of mutation that can occur	Mutation is passed down to gametes
1	mitosis	Polyploidy	✗
2	S phase	Gene mutation	✗
3	meiosis	Aneuploidy	✓
4	S phase	Chromosomal translocation	✓

- A** 1 and 3
B 1, 2 and 3
C 3 and 4
D 1, 2, 3 and 4
- 21** In mice, albinism can be caused by mutation of the tyrosine gene, resulting in replacement of cysteine with serine in the amino acid sequence of the enzyme. No other amino acids change.

The diagram shows the structure of the amino acids cysteine and serine.



Which type of mutation is involved and which type of bond in the enzyme could be affected?

	Type of mutation	Type of bond affected
A	deletion	disulphide bridge
B	deletion	ionic bond
C	substitution	disulphide bridge
D	substitution	ionic bond

- 22** A team of geneticists was examining the inheritance of flower colour and vine length in sweet pea plants. They crossed pure breeding plants that had violet flowers and tall vines with pure breeding plants that had white flowers and dwarf vines. All the F_1 plants were observed to have violet flowers and tall vines. When the F_1 plants were self-pollinated, the results of F_2 generation are shown in the table below.

Phenotypes	Observed number
Violet flower, tall vines	216
Violet flower, dwarf vines	88
White flower, tall vines	67
White flower, dwarf vines	21

A chi-squared test was carried out to determine whether the expected ratio of a dihybrid cross is valid.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

degrees of freedom	probability, P				
	0.50	0.30	0.20	0.05	0.01
1	0.46	1.07	1.64	3.84	6.64
2	1.39	2.41	3.22	5.99	9.21
3	2.37	3.67	4.64	7.82	11.34
4	3.36	4.88	5.99	9.49	13.28

Determine the calculated chi-squared value and the conclusion from the test.

	Chi-squared value	Conclusion
A	4.03	Any deviation from expected ratio is due to chance.
B	4.03	Any deviation from expected ratio is not due to chance.
C	7.82	Any deviation from expected ratio is due to chance.
D	7.82	Any deviation from expected ratio is not due to chance.

- 23** Agouti mice have banded hairs, giving a grey colour. Black mice have unbanded hairs. White mice have no pigment. A cross between a homozygous black mouse and a white mouse produced offspring with agouti hair. Another cross between the black mouse and another white mouse produced some offspring with agouti hair and some with black hair.

What explains these observations on the phenotype of hair of mice?

- A** There is a single gene with two codominant alleles, black and white.
 - B** There is a single gene with three alleles in a dominance series, black → grey → white
 - C** There are two epistatic genes, one controlling pigment production and one controlling banding.
 - D** There are two linked genes, one controlling pigment production and one controlling banding.
- 24** A student made the following claims on photosynthesis:
- Decrease in light intensity results in decrease in rate of Citric acid cycle
 - ATP from substrate level phosphorylation is required for light-independent reaction to take place
 - Photosynthesis involves the conversion of light energy to chemical energy
 - NADP⁺ is oxidised in non-cyclic photophosphorylation
 - Products of cyclic phosphorylation include NADPH and ATP only
 - An action spectrum tells us how much light at different wavelengths is absorbed by a particular photosynthetic pigment.

How many of the student's deductions are correct?

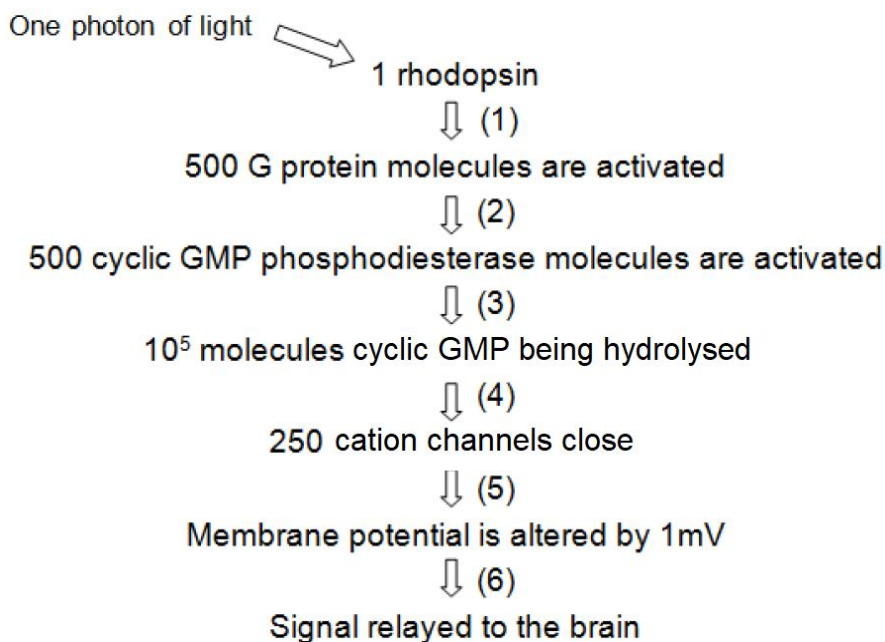
- A** 1
- B** 2
- C** 3
- D** 4

- 25** Yeast cells were obtained and the cell membrane was removed. The contents of the cell were then placed in a test tube. The contents were incubated with NADH without oxygen.

5 cm³ of oxygen was then added and some observations were recorded.

Which is a possible observation?

- A** Carbon dioxide can be detected before and after the addition of oxygen.
 - B** There is an initial rise in pH in the cytosol.
 - C** There is a drop in pH in the mitochondrial intermembrane space after addition of oxygen.
 - D** Ethanol was detected before oxygen was added.
- 26** Rhodopsin is a light sensitive pigment that is present on the rods in the eyes. Rhodopsin is a G protein coupled receptor. The flow chart below shows how the chain of reactions that occur when rhodopsin absorbs a photon of light.



Identify the stages where amplification and cellular responses have occurred.

	Amplification	Cellular Responses
A	1, 3	5
B	1, 3	1, 2, 3, 4, 5, 6
C	1, 2, 3, 4, 5	6
D	2, 3	5, 6

27 Cytochrome c is an essential component in all respiratory electron transport chains. After studying the genetic sequences of a few species, four students made the following conclusions:

- 1 The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
- 2 The sequence of amino acids in cytochrome c is similar in organisms that are from similar environments.
- 3 Most organisms have the same amino acid sequence for cytochrome c.
- 4 When a gene coding for cytochrome c is extracted from a cell of a chimpanzee and inserted into a cell of a cow, the protein expressed is likely able to function in the cell of the cow.

Which conclusions are correct?

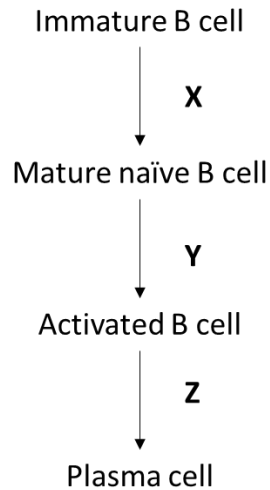
- A** 1 and 3
- B** 2 and 3
- C** 3 and 4
- D** 1, 3 and 4

28 Deer mice are small, ground dwelling rodents. They normally have dark coats. Some paler coated mice have been observed in an area where new sand hills were formed between 8 000 and 15 000 years ago. The sand hills are sparsely covered with scrubby plants. Studies have shown that the change in coat colour was due to a mutation in a gene, causing a single amino acid deletion from the protein for the coat colour pigment. This mutation seems to be spreading through the population.

Which statement most fully explains how the evolution of this species at this site has occurred?

- A** Better camouflage increases the chance of survival by reducing predation.
- B** The occurrence of the mutation provided new variation on which natural selection can act.
- C** The paler colour gives better camouflage against the background of the new sand hills
- D** Deer mice which live longer usually leave more offspring since they have more reproductive opportunities.

- 29** The figure below shows various stages B cells go through before becoming plasma cells.



Which statement is correct?

- A** Class switching occurs during **Y** to change the constant region of the heavy chain.
 - B** Somatic hypermutation occurs during **Z** to give rise to a large repertoire of antigen receptors capable of binding a variety of antigens.
 - C** B cell takes in antigen during **Y** by phagocytosis and presents it to T helper cell.
 - D** Somatic recombination occurs during **X** to change the variable region of the light chain.
- 30** Which is true about climate change?
- A** It is a result of human activities only.
 - B** It is a result of temperature changes.
 - C** It is a result of enhanced greenhouse effects.
 - D** It is a result of ice cap melting and extreme weather patterns.

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Y6 H2 P1 PRELIM

Answers

Qn	Ans	Qn	Ans	Qn	Ans
1	A	11	D	21	C
2	D	12	B	22	A
3	A	13	B	23	C
4	D	14	B	24	A
5	B	15	D	25	C
6	D	16	C	26	A
7	B	17	B	27	C
8	A	18	C	28	B
9	B	19	C	29	D
10	A	20	A	30	C